Abstract

With increasing urban expansion and late-night commuting, personal safety during pedestrian travel remains a critical challenge, particularly for night-shift workers, students, and vulnerable individuals. Traditional navigation systems prioritize travel distance or estimated time of arrival, frequently routing users through unlit, isolated, or high-risk paths. To address this limitation, **SafestWay** (safestway.co.uk) introduces an AI-driven geospatial navigation framework engineered to optimize pedestrian routes based on multi-variable safety metrics.

The application evaluates travel paths by integrating real-time street illumination coverage, proximity to verified 24/7 public assistance nodes (help havens), and active community-reported hazards such as broken lighting or low footfall zones. Built on a modern tech stack utilizing **React.js**, **Vite**, **Tailwind CSS**, and **Leaflet.js / CartoDB** vector mapping, SafestWay offers users interactive route comparisons (*Safest AI Route*, *Balanced Route*, and *Fastest Route*), dynamic safety index scoring (0–100 scale), and an integrated Emergency SOS dispatch hub for real-time GPS coordinate broadcasting.

By shifting the routing paradigm from "fastest path" to "safest path," SafestWay empowers pedestrians with actionable safety intelligence, fostering secure, well-lit, and community-verified night-time transit in urban environments.